STAT300 – Data Analysis Project

(Due: Sunday, November 15, 2026, upload to Canvas by midnight)

This is a group project. That is, you will work on the project in groups, with 3 to 5 students in each group – you choose the group members by yourselves (you may consider choosing group members with different strengths, such as one being good in writing, one being good in R, one being good in data analysis, etc). *The project groups need not be the same as lab groups.* **Each group should work** *independently,* **and should have a leader.** Each group decides the group leader by group members. Each group submits **one** data analysis written report – see details below.

On the cover page of your report, please write the name and student number of each group member, the group leader, and **indicate clearly the roles and contributions each group member** makes in the project. *If you use AI in any part of the project, you must clearly indicate it in your report.*

Each group will need to find a dataset by yourselves (e.g., from internet or books). You should analyze the (same) dataset using at least **three different methods** learned in STAT 300, and then report your results and conclusions. For example, suppose that you wish to compare two groups. You may choose the two-sample t-test, the Wilcoxon test, and the permutation test for the same hypotheses. For each method you choose, you must explain **why** you choose it, what is the **basic idea** behind this method, what is its advantages and limitations, and interpret the results. Finally, you compare the three results with discussions (e.g., why the results differ or agree) and a final conclusion. *You should write your report in a way so that someone who only has an introductory statistics background can understand your methods and are convinced by your conclusions.*

Your report should contain the following materials:

- A cover page: it should include the title of the project, group members (names, student ID numbers, and contributions of each group member), and a short abstract (which summarizes the report).
- Introduction: this section should describe the background and objectives of the study.
- Data description and exploratory data analysis (EDA) : this section should describe where the dataset comes from, the sample size, how the data were collected, etc. EDA may include summary statistics (e.g., means and standard deviations of the variables), graphical displays of the data, and any preliminary conclusions. No assumptions should be made in this section.
- Formal data analysis: this section should include at least three statistical methods for the same dataset, *justifications* of the choices of the methods, *explanations* of the basic idea of the methods in simple languages, *discussions* of their advantages and limitations (e.g., assumptions), *interpretations* of the results, and *comparisons* of the results (e.g., why the

three results agree or dis-agree). Conclusions in this section should be compared with those in EDA and explain any differences.

- Conclusions and discussion: this section should provide overall conclusions, any limitations in the datasets and analyses, and validity of the conclusions in a larger population.
- References: cite any references here.
- Appendix: R code, additional figures/tables, and additional materials.

Requirements:

- Your report is limited to **4 to 6 pages** (12 point fond size), including tables and figures but excluding Appendix and cover page. Your report should be typed (no hand writing).
- You should summarize and interpret your computer outputs, assuming that the readers may not be familiar with R and its output (they may use other software). Do *not* put raw R computer outputs and results in the main text of your report (instead, summarize the key R results in the main text).
- Please put R code, and raw R outputs in the Appendix.
- Each group should work **independently**, and list the **contributions of each group member**.

Rubric:

- **Writing** (20 points): Is the project well written? Is the objective clear? Are the materials well organized?
- **Data and EDA** (20 points): Is the dataset clearly described? Is the EDA done well? Are preliminary conclusions given?
- **Formal analysis** (50 points): Are the choices of the methods justified? Are the ideas of the methods clearly described in a simple way? Are the results interpreted and compared? Are the results presented clearly? Are conclusions clear?
- **Conclusions and discussions** (10 points): Are final conclusions clear and convincing? Are limitations of the data analysis given?

Submissions:

Each group should submit just **one report** (in pdf file) to Canvas by the **group leader**.

File formats:

Please use the following file format: **Stat300-project-Lang-Wu**, where Lang Wu is the name of the group leader.